flight testing, and Alenia Aermacchi contributing to weapon integration, strengthening Europe's technological independence in defense aviation.

Testing and operational milestones

The first flight of the Neuron took place on December 1, 2012, at Istres Air Base, France. Following successful test campaigns in Sweden, Italy, and France, the UCAV demonstrated:

- Autonomous taxi, take off, flight, and landing operations.
- Target approach and payload release simulations.
- Extremely low radar and infrared visibility across multiple radar bands.
- Cooperative data-link compatibility with manned aircraft (Rafale and Gripen).

These results validated the Neuron's potential for integration into future manned–unmanned teaming (MUM-T) and next-generation air dominance programs.

Key specifications:

- *Type*: Unmanned Combat Aerial Vehicle (UCAV) – Technology Demonstrator
- *Developer*: Dassault Aviation, with European partners
- *First flight*: December 1, 2012
- *Wingspan*: 12.5 m (41 ft)
- *Length*: 9.2 m (30 ft)
- *Propulsion*: Rolls-Royce Turbomeca Adour engine
- *Ceiling*: 45,000 ft (13,700 m)
- *Maximum speed*: 980 km/h (610 mph)